Knowledge-Grounded Reasoning and Self-Improvement for Autonomous AI Agents

Brief topic description for Program Officer discussion — ONR FY27 YIP (N0001426SF004)

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Proposed ONR S&T Focus Area: Artificial Intelligence and Autonomy — Intelligent and Collaborative Agents

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Problem. Autonomous AI agents retrieve knowledge, interpret images, call tools, and reason through multi-step plans, yet one unnoticed error can invalidate an otherwise plausible trajectory. Consider an agent reading a photographed inventory sheet for a resupply plan: it extracts the quantity from the neighboring row and passes that number to a planning tool. An outcome reward reveals only that the completed plan failed, while generic self-critique may not locate the first mistake. To reason reliably in unfamiliar settings, an agent needs to assess the evidence for its intermediate steps and decide what to correct.

Central hypothesis. We hypothesize that typed, evidence-linked self-assessment can improve both inference-time recovery and the agent's learned reasoning policy. Agents trained to localize why a step failed should outperform outcome-only reinforcement learning and scalar process reward models on unseen task compositions at comparable computation and tool cost. We will also study when self-assessment should lead to abstention rather than repeated, unproductive retries.

Research plan.

- **Ground each reasoning step in checkable evidence.** Represent intermediate claims and actions together with their source passages, knowledge-base facts, image regions, preconditions, and tool outputs. Build controlled multistep trajectories from entity and event relations, and connect natural agent traces to the evidence they used. The representation must distinguish a wrong observation from an invalid inference or tool action.
- **Learn typed, calibrated self-assessment.** Generalize our FG-PRM synthetic-error methodology from mathematical reasoning to agent trajectories: inject source misreads, entity mismatches, unsupported inferences, wrong tool arguments, and visual attribute or relation errors. Train verifiers to identify the earliest consequential error and its type. Test whether diagnoses trained on synthetic perturbations remain calibrated on natural failures, unfamiliar tools, and new visual inputs.
- **Turn diagnosis into agent self-improvement.** Condition the next action on the diagnosed failure: re-examine an image region, retrieve a missing fact, repair a tool argument, backtrack, or abstain. Use error-aware process feedback to train improved policies, while checking final outcomes independently to detect reward exploitation. Measure improvement across training rounds as well as immediate repair success, including cases where repeated self-correction wastes resources.

Evaluation and expected outcomes. Use MEQA for knowledge-grounded event reasoning, GAIA for multi-step tool and multimodal tasks, and τ-bench for state-changing tool interactions. Since these benchmarks do not provide complete typed error labels for intermediate steps, construct controlled error-injection variants with independently checked outcomes. Measure error localization, self-assessment calibration, successful repair, and task success per tool call. Compare to outcome-only RL, untyped process rewards, and LLM self-critique at matched cost.

Naval relevance. A Naval logistics support agent may combine a photographed inventory list, written guidance, and a resupply-planning tool. If it reads the wrong row, a useful self-assessment identifies the suspect image region and prompts a targeted correction before the tool result is trusted. This addresses the program's interests in uncertain reasoning and self-assessment, visual understanding, and the integration of knowledge, learning, and planning.

Prior work and PI qualifications. Our FG-PRM (Li et al., 2025, Findings of EMNLP) and HiPRAG (Wu et al., 2026, ICLR) develop process-level supervision for reasoning and retrieval. FaithScore (Jing et al., 2024, Findings of EMNLP) and FGAIF (Jing and Du, 2025, TMLR) provide starting points for multimodal feedback. The PI holds a 2024 NSF CAREER award, received a 2023 Amazon Research Award for process-guided fine-tuning, and led system integration for the RESIN-11 team in DARPA KAIROS as a postdoc.